

**Q1. Identify different protocols at Application, Transport and Network Layer.**

- **Application Layer Protocols:** HTTP, DNS, NTP
- **Transport Layer Protocols:** TCP and UDP
- **Network Layer Protocols:** IPv4, IPv6 and ICMP
- In our packet capture, **HTTP** was used at the Application Layer.
- **TCP** was used at the Transport Layer.
- **IPv4** was used at the Network Layer.

**Q2. Find the total data received while opening neverssl.com and cornell.edu.**

- Wireshark was used to capture the network packets.
- The websites **neverssl.com** and **cornell.edu** were opened in the browser.
- The captured traffic was analysed using Wireshark.
- For the analysed TCP Stream **44**, the total captured data was approximately **5 kB**.
- The stream contained **25 packets**.
- The data details were checked using **Statistics → Conversations → TCP**.

**Q3. Analyse the HTTP communication of neverssl.com.**

- **Number of HTTP GET Requests:** 8
- Example request: **GET /online HTTP/1.1**
- **HTTP Version:** HTTP/1.1
- **Application Layer Protocol:** HTTP
- **Transport Layer Protocol:** TCP
- **Network Layer Protocol:** IPv4
- **HTTP Response:** HTTP/1.1 200 OK
- **HTTP Response Content-Length:** 1173 bytes
- **TCP Stream ID:** 44
- **TCP packets in analysed stream:** 25
- **Captured data in analysed stream:** approximately 5 kB
- **TCP Three-Way Handshake:**
  - SYN — Client sends a connection request.
  - SYN-ACK — Server acknowledges the request.
  - ACK — Client acknowledges the server.
- Therefore, the TCP connection was established using **SYN → SYN-ACK → ACK**.

